

SOFTWARE REQUIREMENTS SPECIFICATION

Blood Donation Management Website

Project Title: Blood Donation Website

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1. INTRODUCTION

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements of the Blood Donation Website. The system provides an online platform that connects blood donors with patients, hospitals, and authorized organizations requiring blood.

1.2 Scope

The Blood Donation Website is a web-based application for donor registration, blood-request management, donor searching, notifications, donation history, and administrative monitoring. The system is intended to reduce the time required to locate suitable donors and improve communication between donors and recipients.

1.3 Intended Users

- Donors — people willing to donate blood.
- Recipients — patients or family members seeking blood.
- Hospitals/Organizations — authorized entities requesting or coordinating blood.
- Administrator — manages users, requests, records, and reports.

1.4 Definitions and Acronyms

Term	Description
SRS	Software Requirements Specification
Donor	Person willing to donate blood
Recipient	Person who requires blood
Blood Group	A+, A-, B+, B-, AB+, AB-, O+, or O-
Request	A requirement for a particular blood group
OTP	One-Time Password

2. OVERALL DESCRIPTION

2.1 Product Perspective

The system is a standalone web application accessible through desktop and mobile browsers. It consists of a front-end interface, back-end application, database, authentication mechanism, notification services, and administrator dashboard.

2.2 Product Functions

- User and donor registration/login.
- Donor profile and availability management.
- Creation and management of blood requests.
- Search and matching of donors using blood group and location.
- Notifications and request-status tracking.
- Donation history.
- Administrator management and reporting.

2.3 User Classes

User	Main Capabilities
Donor	Register, manage profile, set availability, view requests, respond to requests, view history.
Recipient	Create blood requests, search suitable donors, track request status.
Hospital	Create and manage requests, search donors, coordinate donations.
Admin	Manage users, requests, records, reports, and system activities.

3. FUNCTIONAL REQUIREMENTS

ID	Requirement	Description
FR-01	User Registration	The system shall allow users to create accounts using name, email, phone, password, location, and u
FR-02	User Login	The system shall authenticate registered users and redirect them to the appropriate dashboard.
FR-03	Donor Registration	The system shall allow users to register as donors and provide blood group, age, location, availability
FR-04	Donor Profile	Donors shall be able to view and update their profile and donation information.
FR-05	Blood Request	Recipients and authorized hospitals shall be able to create blood requests containing blood group, un
FR-06	Search Donors	Users shall be able to search donors by blood group, location, availability, and distance.
FR-07	Search Requests	Donors shall be able to search active blood requests by blood group, location, urgency, and date.
FR-08	Donor Response	Donors shall be able to indicate willingness to donate to a suitable request.
FR-09	Notifications	The system shall notify relevant users when requests are created, accepted, updated, cancelled, or c
FR-10	Donation History	The system shall maintain records of previous donations.
FR-11	Request Status	Requests shall support statuses such as Pending, Active, Donor Found, Completed, and Cancelled.
FR-12	Admin Functions	Administrators shall manage users, verify accounts, manage requests, remove invalid records, and g

4. NON-FUNCTIONAL REQUIREMENTS

Category	Requirement
Performance	Pages and search results should load quickly and support multiple concurrent users.
Security	Passwords must be securely hashed; protected functions require authentication and role-based access.
Usability	The interface should be simple, responsive, mobile-friendly, and provide clear error messages.
Reliability	The system should maintain accurate records and handle failures gracefully.
Availability	The website should be available throughout the day except scheduled maintenance.
Scalability	The architecture should support additional users, hospitals, blood banks, and future features.
Maintainability	The application should use modular components for easy maintenance and upgrades.

5. SYSTEM REQUIREMENTS

5.1 Hardware Requirements

Component	Minimum/Recommended
Client Processor	Dual-core processor or above
Client RAM	2 GB or above
Client Storage	500 MB available space
Server Processor	Quad-core processor or above
Server RAM	8 GB or above
Server Storage	50 GB or above
Connectivity	Stable internet connection

5.2 Software Requirements

Layer	Possible Technologies
Front End	HTML5, CSS3, JavaScript, Bootstrap or React
Back End	Node.js, Python, PHP, or Java
Database	MySQL, PostgreSQL, or MongoDB
Operating System	Windows, Linux, or macOS
Browser	Chrome, Firefox, Edge, or Safari

6. SYSTEM FEATURES

6.1 Home Page

- Website introduction and blood donation awareness.
- Donate Blood, Request Blood, Search Donor, and Login/Register options.
- Emergency blood request section.

6.2 Donor Dashboard

- Donor profile, blood group, availability, active requests, donation history, and notifications.

6.3 Recipient Dashboard

- Create blood request, active requests, matching donors, request status, and notifications.

6.4 Admin Dashboard

- Total users, total donors, active requests, completed donations, pending requests, user management, and reports.

7. USE CASES

Use Case 1: Register as Donor

Actor: Donor

Process: Open website → Register → Select Donor → Enter personal and blood-group details → Submit → System validates → Donor account is created.

Use Case 2: Request Blood

Actor: Recipient/Hospital

Process: Login → Request Blood → Enter patient and hospital details → Select blood group and units → Submit → System creates request → Matching donors are notified.

Use Case 3: Donate Blood

Actor: Donor

Process: Login → View active requests → Select suitable request → Choose "I Can Donate" → System records response → Recipient/Hospital is notified → Donation is coordinated.

8. DATABASE REQUIREMENTS

User Table

Field	Description
User_ID	Unique user ID
Name	User name
Email	User email
Phone	Contact number
Password	Securely stored password hash
Role	Donor / Recipient / Hospital / Admin

Field	Description
Location	User location

Donor Table

Field	Description
Donor_ID	Unique donor ID
User_ID	Associated user
Blood_Group	Donor blood group
Age	Donor age
Gender	Donor gender
Availability	Available / Unavailable
Last_Donation	Previous donation date

Blood Request Table

Field	Description
Request_ID	Unique request ID
User_ID	Requester ID
Blood_Group	Required blood group
Units	Required number of units
Hospital	Hospital name
Location	Hospital location
Urgency	Emergency / Normal
Status	Current request status

Donation Table

Field	Description
Donation_ID	Unique donation ID
Donor_ID	Donor ID
Request_ID	Related request
Donation_Date	Date of donation
Hospital	Donation location
Status	Donation status

9. SYSTEM DIAGRAMS

9.1 System Flowchart

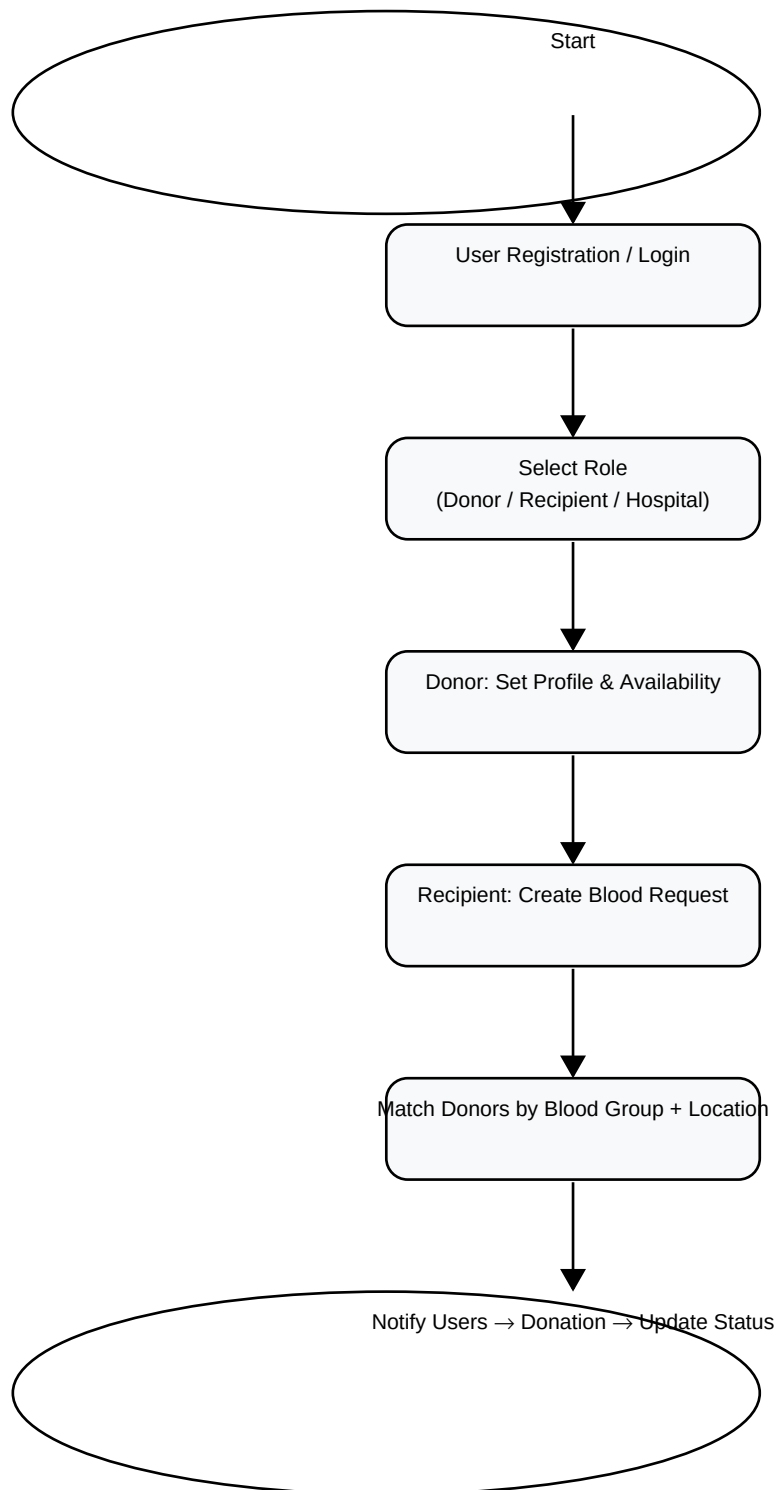


Figure 1. High-level flow of the Blood Donation Website.

9.2 Use-Case Diagram

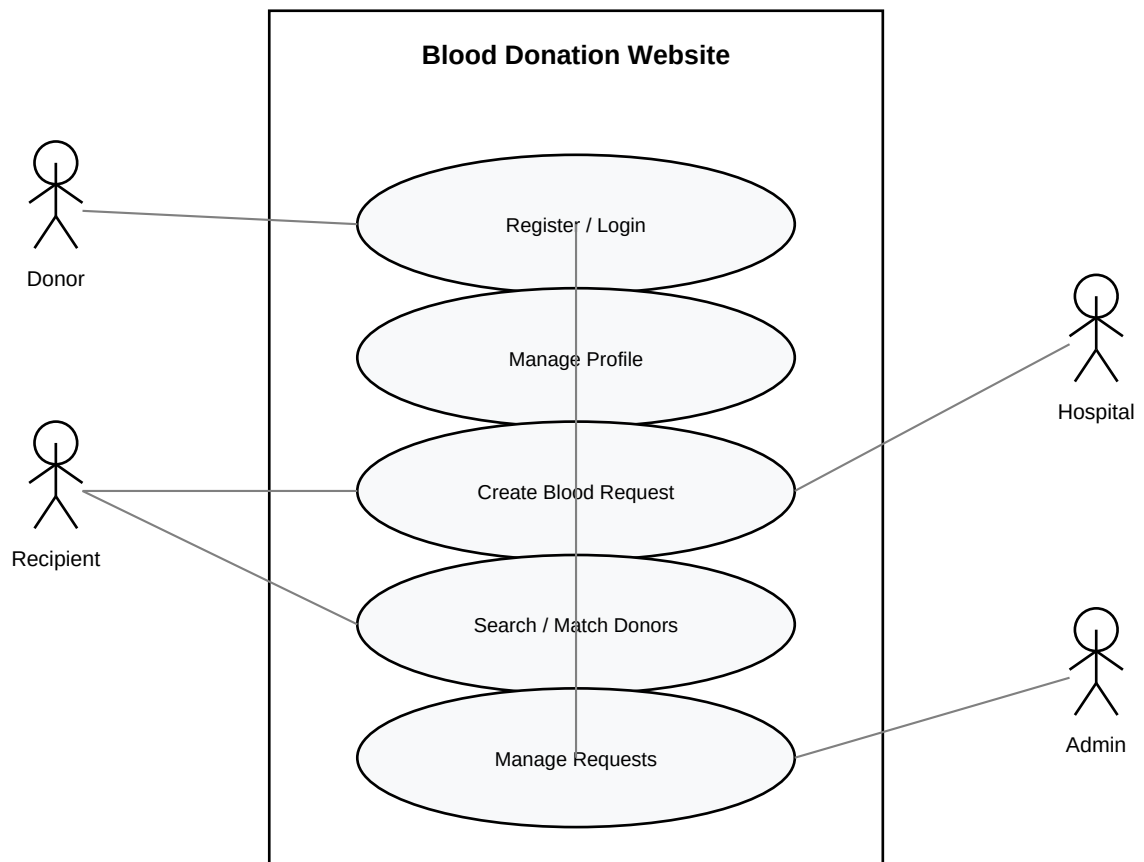


Figure 2. Main actors and interactions with the system.

9.3 Database / ER-Style Diagram

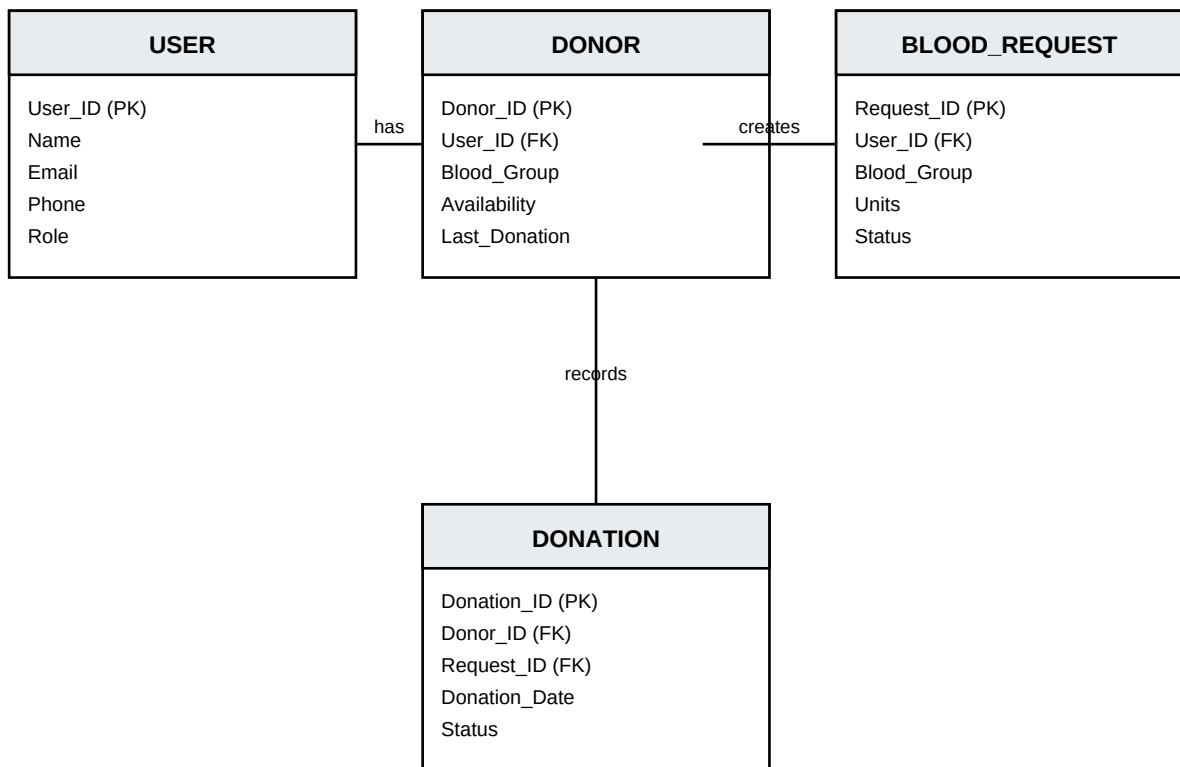


Figure 3. Simplified database relationships.

10. SYSTEM CONSTRAINTS

- Users must have internet access to use the web application.
- Only registered and authorized users can access protected features.
- Donor information must be kept confidential.
- Blood-group information must be entered accurately.
- The website does not determine medical eligibility; qualified medical professionals must make that decision.
- Only authorized administrators can access administrative functions.

11. ASSUMPTIONS AND DEPENDENCIES

- Users provide accurate personal and blood-group information.
- Hospitals and organizations provide valid information.
- Users have a compatible browser and internet connection.
- The database server remains operational.
- Notification services are available when implemented.

12. FUTURE ENHANCEMENTS

- Android and iOS mobile applications.
- GPS-based nearby donor identification.
- AI-assisted donor matching.
- Emergency SOS blood requests.
- SMS and WhatsApp notifications.
- Integration with hospitals and blood banks.
- Digital donor certificates.
- Blood donation eligibility reminders.
- Blood inventory management.
- Multilingual support.
- Online donation appointment booking.
- Analytics dashboard for blood donation statistics.

13. CONCLUSION

The Blood Donation Website provides a centralized and user-friendly platform for connecting blood donors with patients, hospitals, and authorized organizations. The system simplifies donor registration, blood-request creation, donor searching, communication, and request tracking. With secure authentication, structured database management, notifications, and administrative controls, the proposed system can improve the efficiency of blood donation coordination. The architecture also provides a foundation for future enhancements such as mobile applications, GPS-based matching, hospital integration, and emergency notification services.

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